Teacher answer key: Primary 1, Semester 2 (DRAFT)

DRAFT: this answer key is for owner review. Do not give it to pupils. Contact: pypteltd@gmail.com (business email for now, to be changed later)

This key lists the answers and the result of the check on every page. You do not need it for routine marking: a pupil who has a working check has the right answers. To spot-check, compare the check strip with the line "Check result" below.

How the topic tags work

Each entry names the syllabus outcomes the puzzle practises, with the outcome numbers of the Primary 1 content in the Mathematics Syllabus, Primary (2021 edition, pages 31 and 32). The references are for your planning only and are not printed on the pupil pages. "Problem solving" marks a page whose questions come from the kinds found in school tests rather than from one numbered outcome.

Pages in this booklet (Terms 3 and 4)

Strand	Pages
Number and Algebra - Multiplication and Division	S2-01, S2-02, S2-04, S2-05
Number and Algebra - Whole Numbers	S2-03, S2-07, S2-20
Number and Algebra - Addition and Subtraction	S2-06
Measurement and Geometry - Measurement	S2-08, S2-09, S2-14, S2-15, S2-16
Number and Algebra - Money	S2-10, S2-11, S2-12, S2-13
Statistics - Data Representation and Interpretation	S2-17, S2-18, S2-19

The page code in the footer, for example S2-07 (P1-23), gives the page in this booklet and, in brackets, its number in the full set of 40.

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S2-01 (P1-03) Fair Shares 1 star

Topic: Sharing equally and making equal groups (within 20)

Syllabus outcomes: Multiplication and Division 3.1, Multiplication and Division 3.4. Notes on the tag: 3.1, 3.4 (dividing within 20: sharing and grouping, no division sign)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters) and multiplying back.
Answers: 1) 4, 2) 6, 3) 5, 4) 2, 5) 10, 6) 3
Check result: Code word: SHADOW
Teacher note: Pictures of apples, stars, fish and balloons, with empty plates, boxes, bowls and bags. Three items share the objects equally among the containers drawn; three put the objects into groups of a stated size (a dashed loop shows the first group). Every share is exact, with nothing left over. No division sign is used. A pupil can check by multiplying back: containers times each one's share is the number of objects. A printed line says each container gets the same number (the sharer rule).

S2-02 (P1-39) Share and Group 1 star

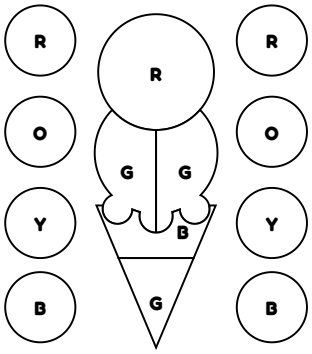
Topic: Sharing and grouping stories within 20, and printed facts

Syllabus outcomes: Multiplication and Division 3.1, Multiplication and Division 3.4 (problem solving). Notes on the tag: 3.1, 3.4 (concepts of multiplication and division, dividing within 20; sharing stories, hidden facts: problem solving)
Puzzle type: Code-breaker (stories). How the pupil checks: Code word (6 letters).
Answers: 1) 3, 2) 8, 3) 4, 4) 6, 5) 5, 6) 2
Check result: Code word: SPONGE
Teacher note: Six stories with pictures. In two the sharer is one of the children: Ali shares 12 sweets with 3 friends, so 4 children share (the page prints that a person who shares also counts). One story groups, one shares among plates, and two give a printed fact (1 car has 4 wheels, 1 bird has 2 legs) and ask how many cars or birds from the wheels or legs counted. Every share is exact and no division sign is used.

S2-03 (P1-14) Colour the Ice Cream 2 stars

Topic: Counting in 2s, 5s and 10s

Syllabus outcomes: Whole Numbers 1.6. Notes on the tag: 1.6 (number patterns); counting in 2s, 5s and 10s leads to multiplication
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and the parts in the same place on the left and on the right match.
Answers: Colour key: 60 = yellow, 65 = red, 70 = blue, 75 = green, 80 = orange
Check result: Tally: yellow 2, red 3, blue 3, green 3, orange 2; the parts in the same place on the left and on the right match.



Teacher note: Patterns count in 2s, 5s or 10s, forwards or back, and the answers are 60 to 80. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); the parts in the same place on the left and on the right match.

S2-04 (P1-05) Times Code 2 stars

Topic: Multiplication within 40: groups of, and the missing number

Syllabus outcomes: Multiplication and Division 3.1, Multiplication and Division 3.2, Multiplication and Division 3.3. Notes on the tag: 3.1 to 3.3 (concept of multiplication, sign x, multiplying within 40)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters) and putting the number back.
Answers: 1) 4, 2) 15, 3) 12, 4) 2, 5) 6, 6) 16
Check result: Code word: SQUARE
Teacher note: Second multiplication page (the equal groups page uses bags). Every item has an array of objects: the rows are the groups and the objects in a row are the number in each group, so a pupil can count. Two items give the product with the sign, two use the words groups of, and two have a missing number in a multiplication sentence, which is the two-star step: the pupil must undo the sum. The number of groups is 2, 3 or 4, the number in a group is 2 to 5, and every product is within 20.

S2-05 (P1-40) True or False 2 stars

Topic: Multiplication and sharing: true or false statements

Syllabus outcomes: Multiplication and Division 3.1, Multiplication and Division 3.2, Multiplication and Division 3.3. Notes on the tag: 3.1, 3.2, 3.3 (concepts of multiplication, use of x, multiplying within 40; true or false: reasoning)
Puzzle type: True or false table + code word. How the pupil checks: The letter beside the ticked T or F in each row, read down the page, spells a word (6 letters).
Answers: 1) F (T), 2) F (E), 3) T (A), 4) F (C), 5) T (U), 6) T (P)
Check result: Letters in order: TEACUP
Teacher note: Six statements about groups, the sign x and sharing, three true and three false, within 40 (sharing within 20). A false statement is a common slip: adding instead of multiplying, or one group too many or too few. Each choice has a letter. The right choices spell the word; every other way of ticking was tested against a very large word list and none spells a word. No division sign is used.

S2-06 (P1-11) Big Number Sums 2 stars

Topic: Add and subtract within 100

Syllabus outcomes: Addition and Subtraction 2.5, Addition and Subtraction 2.7. Notes on the tag: 2.5, 2.7 (add and subtract within 100: 2-digit and ones, 2-digit and tens, missing number)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters) and putting the number back.
Answers: 1) 21, 2) 81, 3) 94, 4) 30, 5) 17, 6) 88
Check result: Code word: FAMILY
Teacher note: Numbers are within 100 and no sum crosses a ten (no renaming). Items: 2-digit plus ones, 2-digit plus tens, take away ones, take away tens, and two missing-number items. The missing-number items are the two-star step: the pupil must undo the sum.

S2-07 (P1-23) Stack It Up 1 star

Topic: Adding and subtracting tens and ones, written in boxes (within 100)

Syllabus outcomes: Addition and Subtraction 2.5, Addition and Subtraction 2.6. Notes on the tag: 2.5, 2.6 (adding and subtracting within 100 using algorithms; tens and ones, no renaming)
Puzzle type: Code-breaker (written sums). How the pupil checks: Code word (6 letters).
Answers: 1) 64 - 32 = 32, 2) 21 + 44 = 65, 3) 56 + 13 = 69 (missing digit 6), 4) 75 - 35 = 40, 5) 89 - 39 = 50 (missing digit 9), 6) 35 + 13 = 48
Check result: Code word: DONKEY
Teacher note: Four written sums (two adding, two taking away) have two boxes for the answer, tens then ones. Two sums have the answer printed and a box for the ones of the top number, so the pupil must undo the sum. All sums are within 100 and none needs renaming: the ones always add to 9 or less and a take-away never needs a borrow. The syllabus lists algorithms (written methods) for adding and subtracting within 100 without giving a limit; the safe reading used here is no renaming.

S2-08 (P1-32) Measure and Draw 1 star

Topic: Length in cm: measure a bar and draw a line

Syllabus outcomes: Length 1.1, Length 1.2, Length 1.4. Notes on the tag: Length 1.1, 1.2, 1.4 (measuring length in cm; abbreviation cm; measuring and drawing a line segment to the nearest cm)
Puzzle type: Code-breaker (measure and draw). How the pupil checks: Code word (6 letters).
Answers: 1) line 3 cm = S, 2) line 2 cm = T, 3) line 12 cm = R, 4) bar from 2 to 8: 6 cm = I, 5) bar from 3 to 7: 4 cm = N, 6) bar from 2 to 10: 8 cm = G
Check result: Code word: STRING
Teacher note: Three bars start at 2 or 3 on the ruler (not at 0) and are measured in cm. Three lines are drawn from a dot at 0 to a given length. On the drawing rulers each cm number has its code letter under it, so the letter at the end of the line is the answer; on the measuring rulers the pupil uses the code table. The page cannot see the line the pupil draws: it checks only that the pupil finds the right end mark. Every length is a whole number of cm (to the nearest cm).

S2-09 (P1-06) What Time Is It? 1 star

Topic: Time: o'clock and half past

Syllabus outcomes: Time 2.1. Notes on the tag: Time 2.1 (tell time: o'clock and half past)
Puzzle type: Code-breaker. How the pupil checks: Code word (6 letters).
Answers: 1) 8:00 (8 o'clock), 2) 4:00 (4 o'clock), 3) 9:00 (9 o'clock), 4) 1:30 (half past 1), 5) 11:30 (half past 11), 6) 2:30 (half past 2)
Check result: Code word: SECOND
Teacher note: Three o'clock times and three half past times; the short hand is never close to the long hand. The code table holds the six right times and four wrong readings (the hour one too late, the hour one too early, o'clock read as half past, or the other way round). Times to 5 minutes are on the Clock Time page.

S2-10 (P1-15) Coin Purses 1 star

Topic: Money: counting coins in cents (up to $1)

Syllabus outcomes: Money 1.1. Notes on the tag: Money 1.1 (count money in cents up to $1)
Puzzle type: Code-breaker. How the pupil checks: Code word (5 letters).
Answers: 1) 80¢, 2) 60¢, 3) 40¢, 4) 25¢, 5) 95¢
Check result: Code word: PIANO
Teacher note: Coins used: 5¢, 10¢, 20¢, 50¢. Each coin shows its value and the word cents. The coins are original line drawings (a small picture on each, not a copy of any real coin), with sizes in the true proportion of the real coins. Totals stay below $1.

S2-11 (P1-28) Fewest Coins 1 star

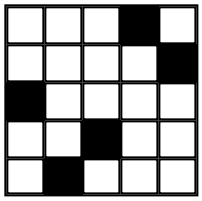
Topic: Money: counting coins in cents; fewest coins and ways to pay

Syllabus outcomes: Money 1.1 (problem solving). Notes on the tag: Money 1.1 (counting amount of money in cents up to $1; fewest coins and ways to pay come from school tests: problem solving)
Puzzle type: Code-breaker (coins). How the pupil checks: Code word (5 letters).
Answers: 1) 2, 2) 3, 3) 9, 4) 4, 5) 1 (ticked purses: 2)
Check result: Code word: STAMP
Teacher note: Coins are 5¢, 10¢, 20¢ and 50¢ (original drawings with the value and the word cents). Two items ask for the fewest coins, two ask how many of one coin make an amount, and one asks the pupil to tick the purses that hold exactly the amount and say how many. Fewest coins and ways to pay come from the kinds of question in school tests; the syllabus outcome is counting money.

S2-12 (P1-16) Shop Prices 2 stars

Topic: Money: dollars, more and less

Syllabus outcomes: Money 1.1, Addition and Subtraction 2.7 (problem solving). Notes on the tag: Money 1.1 (dollars up to $100), 2.7 (add, subtract); guess and check, problem solving
Puzzle type: Logic grid + total. How the pupil checks: Letters beside the prices, then a code letter for the total, spell a word (6 letters).
Answers: pen: $12 (F), book: $15 (R), ball: $5 (O), kite: $8 (Z), cup: $7 (E); the ball and the pen cost $17 together (code letter N)
Check result: Letters in order: FROZEN



Teacher note: Five items, five prices, each price with a letter. The page shows a worked example of crosses and ticks. '$3 more than' means a price $3 higher: pupils test each price against it. The last answer is added from the grid and looked up in a small code table, so a wrong grid or a wrong sum changes the word.

S2-13 (P1-27) Which Two Items? 2 stars

Topic: Money in dollars: which two things cost exactly the amount paid

Syllabus outcomes: Money 1.1, Addition and Subtraction 2.5 (problem solving). Notes on the tag: Money 1.1 (dollars up to $100), 2.5 (adding within 100); guess and check, problem solving
Puzzle type: Logic (price board). How the pupil checks: The letters of the two things each buyer pays for, cheaper thing first, spell a word (6 letters).
Answers: Jenny pays $22: toy car $9 (M) and kite $13 (A); Aisha pays $32: cup $8 (G) and boat $24 (N); Lily pays $34: ball $16 (E) and drum $18 (T)
Check result: Letters in order: MAGNET
Teacher note: Six things with prices in dollars. Each of three people pays an exact amount for two of them; for each amount only one pair of things works, and the three pairs use all six things. The pupil tries pairs (guess and check), then writes the letter of the cheaper thing and then the dearer one. The six letters have exactly one arrangement that is a word, and every single wrong pair was tested: none spells a word. Prices are whole dollars within $100.

S2-14 (P1-18) Clock Time 2 stars

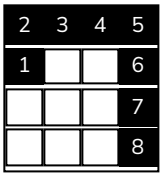
Topic: Time to 5 minutes: 1 hour and half an hour later

Syllabus outcomes: Time 2.1, Time 2.4. Notes on the tag: Time 2.1 and 2.4 (tell time to 5 minutes; durations of 1 hour or half an hour)
Puzzle type: Code-breaker. How the pupil checks: Code word (5 letters).
Answers: 1) shows 8:10, 1 hour later 9:10, 2) shows 1:25, half an hour later 1:55, 3) shows 1:30 (half past 1), 1 hour later 2:30 (half past 2), 4) shows 11:25, half an hour later 11:55, 5) shows 11:00 (11 o'clock), 1 hour later 12:00 (12 o'clock)
Check result: Code word: CLIMB
Teacher note: Times to 5 minutes are usually taught late in Primary 1 (use the What Time Is It? page for o'clock and half past). Each item is two steps: read the clock, then add 1 hour or half an hour. None of the half-hour additions cross the hour. The decoy times in the table are the hour one too late, the hands read the wrong way round. The hands are never closer than 30 degrees. a.m. and p.m. are not used on this page.

S2-15 (P1-30) Morning or Evening? 1 star

Topic: Time: am and pm with times to 5 minutes

Syllabus outcomes: Time 2.1, Time 2.2. Notes on the tag: Time 2.1, 2.2 (telling time to 5 minutes; use of am and pm)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, and every box must make sense (the right am or pm for the event).
Answers: Route (row, column from top-left): (2,1) 9:00 pm bedtime -> (1,1) 8:00 pm dinner -> (1,2) 9:00 am breakfast -> (1,3) 8:00 am school starts -> (1,4) 4:00 pm play outside -> (2,4) 7:00 am wake up -> (3,4) 7:30 am wake up -> (4,4) 5:00 pm play outside
Check result: Route reaches FINISH through 8 boxes; every box is a sensible time for its event.



Teacher note: A 4 by 4 grid of times with an event, for example a time for breakfast. am is the morning and pm is the afternoon and evening; times are written in digits with am or pm and no full stops. 10 boxes make sense and 6 do not (the same time with the wrong am or pm, or a time that does not fit the event). 2 sensible boxes beside the route lead nowhere, so the box count and FINISH catch a wrong turn.

S2-16 (P1-29) How Long Does It Last? 1 star

Topic: Time: how long a thing lasts (1 h or 30 min) and when it ends

Syllabus outcomes: Time 2.3, Time 2.4, Time 2.1. Notes on the tag: Time 2.1, 2.3, 2.4 (telling time to 5 minutes; abbreviations h and min; duration of one hour or half hour)
Puzzle type: Code-breaker (clocks and a timetable). How the pupil checks: Code word (6 letters).
Answers: 1) 1 h (starts 4:45, ends 5:45), 2) 30 min (starts 7:45, ends 8:15), 3) 12:15, 4) 2:05, 5) 8:15, 6) 12:10
Check result: Code word: SAILOR
Teacher note: Times are read to 5 minutes and written with digits. Durations are always 1 h (one hour) or 30 min (half an hour); h and min are the syllabus abbreviations and are used on this page only. Questions 1 and 2 give the start and end clocks and ask how long. In the show table, two rows ask for an end time (add 1 h or 30 min) and two ask for a start time (go back 1 h or 30 min). A 30 min step may pass the next whole hour. The table's decoys are the usual slips: adding when the pupil should go back, or using the wrong length.

S2-17 (P1-35) Read the Picture Graph 1 star

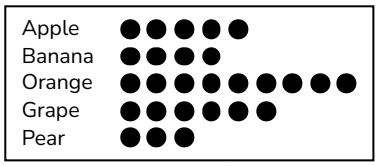
Topic: Picture graphs: read the graph and answer questions

Syllabus outcomes: Picture Graphs 1.1. Notes on the tag: Statistics 1.1 (reading and interpreting data from picture graphs; one picture stands for one child)
Puzzle type: Code-breaker (picture graph). How the pupil checks: Code word (6 letters).
Answers: 1) Cat, 2) Fish, 3) 5, 4) 9, 5) 6, 6) 26
Check result: Code word: TURKEY
Teacher note: A finished graph with one face for one child (never a scale) and five rows. Two questions ask which pet (most, fewest) and four ask for a number: one row, two rows together, how many more, and the total. The code table has pet names and numbers, so a pupil who writes a number where a pet is wanted gets no letter. The total of all rows is read from the graph, not given.

S2-18 (P1-20) Fruit Graph 2 stars

Topic: Picture graphs: read, complete and answer

Syllabus outcomes: Picture Graphs 1.1. Notes on the tag: Statistics 1.1 (read and interpret picture graphs)
Puzzle type: Graph detective. How the pupil checks: Two sums: the total of all the faces, and the row chosen in a) plus the answer to b) (neither is used to find any row).
Answers: Apple: 5, Banana: 4, Orange: 9, Grape: 6, Pear: 3; a) Orange; b) 2
Check result: Total 27 faces; second sum 9 + 2 = 11

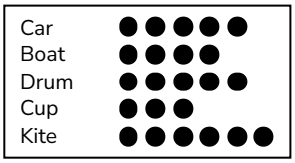


Teacher note: Each smiley face stands for one child (a scale of 1: one picture symbol for one child). A pupil who draws a plain circle instead of a face counts the same, so the checks do not change. The three clues fix the three missing rows one by one. The total of 27 is printed only in the check strip, so it is a separate test: a pupil who misreads a clue or draws one circle too many or too few gets a different total. The second sum tests a) and b): it is 11 only when the right row is chosen and b) is right. The three clues start from Apple or Banana; a pupil who starts a clue from the wrong one of the two, or who reads more as fewer, also gets a wrong total or a wrong second sum (every such mix was tried). Question b) names two other fruits, so it does not give away the answer to a). More and fewer are not mirror images here (the Grape clue is 2 and the Pear clue is 1), and the first two clues add different amounts (4 and 2), so a pupil who swaps the two words or swaps Apple and Banana gets a different total or a different second sum.

S2-19 (P1-36) Graph Detective Two 2 stars

Topic: Picture graphs: read, complete and answer from clues

Syllabus outcomes: Picture Graphs 1.1. Notes on the tag: Statistics 1.1 (reading and interpreting data from picture graphs; completing a graph from clues)
Puzzle type: Graph detective. How the pupil checks: Two sums: the total of all the stars, and the row chosen in a) plus the answer to b) (neither is used to find any row).
Answers: Car: 5, Boat: 4, Drum: 5, Cup: 3, Kite: 6; a) Kite; b) 2
Check result: Total 23 stars; second sum 6 + 2 = 8



Teacher note: Each star stands for one child (one picture for one child, no scale). Two rows are drawn and three are found from three clues: one says two rows have the same, one gives the total of two rows, and one gives a difference between two rows. The total of 23 is printed only in the check strip, so it is a separate test, not a clue. The second sum tests the answers to a) and b). All the ways of misreading the three clues (the wrong row used, the total read as one row, more and fewer exchanged) were tried; every one fails the total or the second sum.

S2-20 (P1-22) Two-Step Stories 2 stars

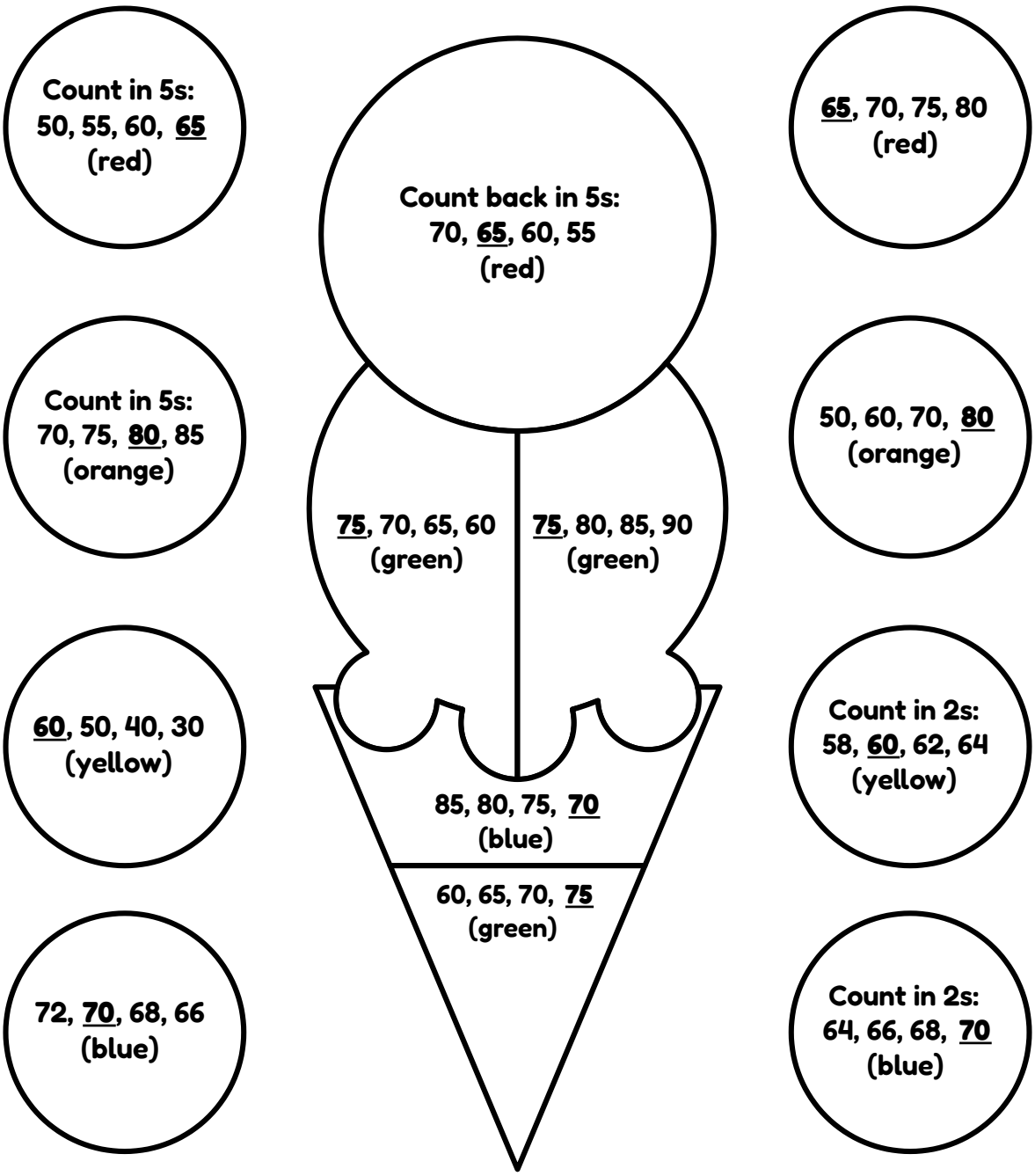
Topic: Two-step stories: add, take away, work backwards

Syllabus outcomes: Addition and Subtraction 2.1, Addition and Subtraction 2.2, Addition and Subtraction 2.5, Addition and Subtraction 2.7 (problem solving). Notes on the tag: 2.1, 2.2, 2.5, 2.7 (two-step stories within 100: problem solving, work backwards, a printed fact)
Puzzle type: Code-breaker (two-step stories). How the pupil checks: Code word (6 letters).
Answers: 1) 43 then 35, 2) 16 then 14, 3) 4 then 7, 4) 14 then 13, 5) 85 then 83, 6) 12 then 9
Check result: Code word: GUITAR
Teacher note: Six stories with two boxes each. The second box (the end answer) is the one looked up in the code table. Two stories say 'at first': the pupil starts from the number now and works backwards. One story has a printed fact (1 tricycle has 3 wheels) so a pupil is not stopped by a fact they may not know. Two stories use tens (no renaming). The first box is not looked up in the table; a wrong first answer almost always makes the end answer wrong. Pupils can put the end answer back into the story to check it makes sense.

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Answers: S2-03 (P1-14) Colour the Ice Cream

Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 60 = yellow, 65 = red, 70 = blue, 75 = green, 80 = orange.



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